Choice and well-ordering are equivalent with explicit universe control

The Univalent Foundations Program

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Editorial note. Complete fixed-author source spans retained; original CC BY-SA attribution and macros reused. Supporting statements do not create extra counted problems.

Editorial note. Construct compatible well-founded choice enumerations across ordinals and prove that the image exhausts the set by a universe-correct ordinal bound and successor argument. All local ordinal simulation and union constructions are included.

Editorial note. The original fixed-edition label-number file does not list rmk:quotient-of-non-set. Its actual four-line author remark is included with the native label instead of assigning an unverified printed number.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 5.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1660–1660. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 10.1.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1670–1670. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

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Source reference. Original reference 5.3: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1724–1724. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 3.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1735–1735. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Remark 10.0.1. It is not actually necessary for the definition of set-quotients, and most of their properties, that A be a set. However, this is generally the case of most interest.

10.1 Ordinal numbers

Definition 10.1.1. Let A be a set and

$$(- < -) : A \to A \to \mathsf{Prop}$$

a binary relation on A. We define by induction what it means for an element $a : A$ to be **accessible** by $<$:

- If b is accessible for every $b < a$, then a is accessible.

We write $\mathsf{acc}(a)$ to mean that a is accessible.

It may seem that such an inductive definition can never get off the ground, but of course if a has the property that there are *no* b such that $b < a$, then a is vacuously accessible.

Note that this is an inductive definition of a family of types, like the type of vectors considered in original reference 5.7. More precisely, it has one constructor, say $\mathsf{acc}_<$, with type

$$\mathsf{acc}_< : \prod_{a:A} \Big(\prod_{b:A} (b < a) \to \mathsf{acc}(b) \Big) \to \mathsf{acc}(a).$$

The induction principle for acc says that for any $P : \prod_{(a:A)} \mathsf{acc}(a) \to \mathcal{U}$, if we have

$$f : \prod_{(a:A)} \prod_{(h:\prod_{(b:A)}(b<a)\to\mathsf{acc}(b))} \Big(\prod_{(b:A)} \prod_{(l:b<a)} P(b, h(b,l)) \Big) \to P(a, \mathsf{acc}_<(a,h)),$$

then we have $g : \prod_{(a:A)} \prod_{(c:\mathsf{acc}(a))} P(a,c)$ defined by induction, with

$$g(a, \mathsf{acc}_<(a,h)) \equiv f(a,\ h,\ \lambda b.\, \lambda l.\, g(b, h(b,l))).$$

This is a mouthful, but generally we apply it only in the simpler case where $P : A \to \mathcal{U}$ depends only on A. In this case the second and third arguments of f may be combined, so that what we have to prove is

$$f : \prod_{a:A} \Big(\prod_{b:A} (b < a) \to \mathsf{acc}(b) \times P(b) \Big) \to P(a).$$

That is, we assume every $b < a$ is accessible and $g(b) : P(b)$ is defined, and from these define $g(a) : P(a)$.

The omission of the second argument of P is justified by the following lemma, whose proof is the only place where we use the more general form of the induction principle.

Lemma 10.1.2. *Accessibility is a mere property.*

Proof. We must show that for any $a : A$ and $s_1, s_2 : \mathsf{acc}(a)$ we have $s_1 = s_2$. We prove this by induction on s_1, with

$$P_1(a, s_1) :\equiv \prod_{s_2 : \mathsf{acc}(a)} (s_1 = s_2).$$

Thus, we must show that for any $a : A$ and $h_1 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$ and

$$k_1 : \prod_{(b:A)} \prod_{(l : b<a)} \prod_{(t:\mathsf{acc}(b))} h_1(b, l) = t,$$

we have $\mathsf{acc}_<(a, h) = s_2$ for any $s_2 : \mathsf{acc}(a)$. We regard this statement as $\prod_{(a:A)} \prod_{(s_2:\mathsf{acc}(a))} P_2(a, s_2)$, where

$$P_2(a, s_2) :\equiv \prod_{(h_1 : \cdots)} \prod_{(k_1 : \cdots)} (\mathsf{acc}_<(a, h_1) = s_2);$$

thus we may prove it by induction on s_2. Therefore, we assume $h_2 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$, and k_2 with a monstrous but irrelevant type, and must show that for any h_1 and k_1 with types as above, we have $\mathsf{acc}_<(a, h_1) = \mathsf{acc}_<(a, h_2)$. By function extensionality, it suffices to show $h_1(b, l) = h_2(b, l)$ for all $b : A$ and $l : b < a$. This follows from k_1. □

Definition 10.1.3. A binary relation $<$ on a set A is **well-founded** if every element of A is accessible.

The point of well-foundedness is that for $P : A \to \mathcal{U}$, we can use the induction principle of acc to conclude $\prod_{(a:A)} \mathsf{acc}(a) \to P(a)$, and then apply well-foundedness to conclude $\prod_{(a:A)} P(a)$. In other words, if from $\forall(b : A).\, (b < a) \to P(b)$ we can prove $P(a)$, then $\forall(a : A).\, P(a)$. This is called **well-founded induction**.

Lemma 10.1.4. *Well-foundedness is a mere property.*

Proof. Well-foundedness of $<$ is the type $\prod_{(a:A)} \mathsf{acc}(a)$, which is a mere proposition since each $\mathsf{acc}(a)$ is. □

Example 10.1.5. Perhaps the most familiar well-founded relation is the usual strict ordering on $\mathbb{N}$. To show that this is well-founded, we must show that n is accessible for each $n : \mathbb{N}$. This is just the usual proof of "strong induction" from ordinary induction on $\mathbb{N}$.

Specifically, we prove by induction on $n : \mathbb{N}$ that k is accessible for all $k \leq n$. The base case is just that 0 is accessible, which is vacuously true since nothing is strictly less than 0. For the inductive step, we assume that k is accessible for

all $k \leq n$, which is to say for all $k < n + 1$; hence by definition $n + 1$ is also accessible.

A different relation on $\mathbb{N}$ which is also well-founded is obtained by setting only $n < \mathsf{succ}(n)$ for all $n : \mathbb{N}$. Well-foundedness of this relation is almost exactly the ordinary induction principle of $\mathbb{N}$.

Example 10.1.6. Let $A : \mathsf{Set}$ and $B : A \to \mathsf{Set}$ be a family of sets. Recall from original reference 5.3 that the W-type $\mathsf{W}_{(a:A)} B(a)$ is inductively generated by the single constructor

- $\mathsf{sup} : \prod_{(a:A)} (B(a) \to \mathsf{W}_{(x:A)} B(x)) \to \mathsf{W}_{(x:A)} B(x)$

We define the relation $<$ on $\mathsf{W}_{(x:A)} B(x)$ by recursion on its second argument:

- For any $a : A$ and $f : B(a) \to \mathsf{W}_{(x:A)} B(x)$, we define $w < \mathsf{sup}(a, f)$ to mean that there merely exists a $b : B(a)$ such that $w = f(b)$.

Now we prove that every $w : \mathsf{W}_{(x:A)} B(x)$ is accessible for this relation, using the usual induction principle for $\mathsf{W}_{(x:A)} B(x)$. This means we assume given $a : A$ and $f : B(a) \to \mathsf{W}_{(x:A)} B(x)$, and also a lifting $f' : \prod_{(b:B(a))} \mathsf{acc}(f(b))$. But then by definition of $<$, we have $\mathsf{acc}(w)$ for all $w < \mathsf{sup}(a, f)$; hence $\mathsf{sup}(a, f)$ is accessible.

Well-foundedness allows us to define functions by recursion and prove statements by induction, such as for instance the following. Recall from original reference 3.5 that $\mathcal{P}(B)$ denotes the *power set* $\mathcal{P}(B) :\equiv (B \to \mathsf{Prop})$.

Lemma 10.1.7. *Suppose B is a set and we have a function*

$$g : \mathcal{P}(B) \to B$$

Then if $<$ is a well-founded relation on A, there is a function $f : A \to B$ such that for all $a : A$ we have

$$f(a) = g\Big(\{\, f(a') \mid a' < a \,\} \Big).$$

(We are using the notation for images of subsets from original reference 10.1.2.)

Proof. We first define, for every $a : A$ and $s : \mathsf{acc}(a)$, an element $\bar{f}(a, s) : B$. By induction, it suffices to assume that s is a function assigning to each $a' < a$ a witness $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have an element $\bar{f}(a', s(a')) : B$. In this case, we define

$$\bar{f}(a, s) :\equiv g\Big(\{\, \bar{f}(a', s(a')) \mid a' < a \,\} \Big).$$

Now since $<$ is well-founded, we have a function $w : \prod_{(a:A)} \mathsf{acc}(a)$. Thus, we can define $f(a) :\equiv \bar{f}(a, w(a))$. □

In classical logic, well-foundedness has a more well-known reformulation. In the following, we say that a subset $B : \mathcal{P}(A)$ is **nonempty** if it is unequal to the empty subset $(\lambda x. \bot) : \mathcal{P}(X)$. We leave it to the reader to verify that assuming excluded middle, this is equivalent to mere inhabitation, i.e. to the condition $\exists (x : A).\, x \in B$.

Lemma 10.1.8. *Assuming excluded middle, $<$ is well-founded if and only if every nonempty subset $B : \mathcal{P}(A)$ merely has a minimal element.*

Proof. Suppose first $<$ is well-founded, and suppose $B \subseteq A$ is a subset with no minimal element. That is, for any $a : A$ with $a \in B$, there merely exists a $b : A$ with $b < a$ and $b \in B$.

We claim that for any $a : A$ and $s : \mathsf{acc}(a)$, we have $a \notin B$. By induction, we may assume s is a function assigning to each $a' < a$ a proof $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have $a' \notin B$. If $a \in B$, then by assumption, there would merely exist a $b < a$ with $b \in B$, which contradicts this assumption. Thus, $a \notin B$; this completes the induction. Since $<$ is well-founded, we have $a \notin B$ for all $a : A$, i.e. B is empty.

Now suppose each nonempty subset merely has a minimal element. Let $B = \{\, a : A \mid \neg\mathsf{acc}(a) \,\}$. Then if B is nonempty, it merely has a minimal element. Thus there merely exists an $a : A$ with $a \in B$ such that for all $b < a$, we have $\mathsf{acc}(b)$. But then by definition (and induction on truncation), a is merely accessible, and hence accessible, contradicting $a \in B$. Thus, B is empty, so $<$ is well-founded. □

Definition 10.1.9. A well-founded relation $<$ on a set A is **extensional** if for any $a, b : A$, we have

$$\Big(\forall(c : A).\, (c < a) \Leftrightarrow (c < b)\Big) \to (a = b).$$

Note that since A is a set, extensionality is a mere proposition. This notion of "extensionality" is unrelated to function extensionality, and also unrelated to the extensionality of identity types. Rather, it is a "local" counterpart of the axiom of extensionality in classical set theory.

Theorem 10.1.10. *The type of extensional well-founded relations is a set.*

Proof. By the univalence axiom, it suffices to show that if $(A, <)$ is extensional and well-founded and $f : (A, <) \cong (A, <)$, then $f = \mathsf{id}_A$. We prove by induction on $<$ that $f(a) = a$ for all $a : A$. The inductive hypothesis is that for all $a' < a$, we have $f(a') = a'$.

Now since A is extensional, to conclude $f(a) = a$ it is sufficient to show

$$\forall(c : A).\, (c < f(a)) \Leftrightarrow (c < a).$$

However, since f is an automorphism, we have $(c < a) \Leftrightarrow (f(c) < f(a))$. But $c < a$ implies $f(c) = c$ by the inductive hypothesis, so $(c < a) \to (c < f(a))$. On the other hand, if $c < f(a)$, then $f^{-1}(c) < a$, and so $c = f(f^{-1}(c)) = f^{-1}(c)$ by the inductive hypothesis again; thus $c < a$. Therefore, we have $(c < a) \Leftrightarrow (c < f(a))$ for any $c : A$, so $f(a) = a$. □

Definition 10.1.11. If $(A, <)$ and $(B, <)$ are extensional and well-founded, a **simulation** is a function $f : A \to B$ such that

(i) if $a < a'$, then $f(a) < f(a')$, and
(ii) for all $a : A$ and $b : B$, if $b < f(a)$, then there merely exists an $a' < a$ with $f(a') = b$.

Lemma 10.1.12. *Any simulation is injective.*

Proof. We prove by double well-founded induction that for any $a, b : A$, if $f(a) = f(b)$ then $a = b$. The inductive hypothesis for $a : A$ says that for any $a' < a$, and any $b : B$, if $f(a') = f(b)$ then $a = b$. The inner inductive hypothesis for $b : A$ says that for any $b' < b$, if $f(a) = f(b')$ then $a = b'$.

Suppose $f(a) = f(b)$; we must show $a = b$. By extensionality, it suffices to show that for any $c : A$ we have $(c < a) \Leftrightarrow (c < b)$. If $c < a$, then $f(c) < f(a)$ by Definition 10.1.11(i). Hence $f(c) < f(b)$, so by Definition 10.1.11(ii) there merely exists $c' : A$ with $c' < b$ and $f(c) = f(c')$. By the inductive hypothesis for a, we have $c = c'$, hence $c < b$. The dual argument is symmetrical. □

In particular, this implies that in Definition 10.1.11(ii) the word "merely" could be omitted without change of sense.

Corollary 10.1.13. *If $f : A \to B$ is a simulation, then for all $a : A$ and $b : B$, if $b < f(a)$, there* purely *exists an $a' < a$ with $f(a') = b$.*

Proof. Since f is injective, $\sum_{(a:A)}(f(a) = b)$ is a mere proposition. □

We say that a subset $C : \mathcal{P}(B)$ is an **initial segment** if $c \in C$ and $b < c$ imply $b \in C$. The image of a simulation must be an initial segment, while the inclusion of any initial segment is a simulation. Thus, by univalence, every simulation $A \to B$ is *equal* to the inclusion of some initial segment of B.

Theorem 10.1.14. *For a set A, let $P(A)$ be the type of extensional well-founded relations on A. If $<_A : P(A)$ and $<_B : P(B)$ and $f : A \to B$, let $H_{<_A <_B}(f)$ be the mere proposition that f is a simulation. Then (P, H) is a standard notion of structure over $\mathcal{S}et$ in the sense of original reference 9.8.*

Proof. We leave it to the reader to verify that identities are simulations, and that composites of simulations are simulations. Thus, we have a notion of structure. For standardness, we must show that if $<$ and $\prec$ are two extensional well-founded relations on A, and id_A is a simulation in both directions, then $<$ and $\prec$ are equal. Since extensionality and well-foundedness are mere propositions, for this it suffices to have $\forall(a, b : A).\, (a < b) \Leftrightarrow (a \prec b)$. But this follows from Definition 10.1.11(i) for id_A. □

Corollary 10.1.15. *There is a category whose objects are sets equipped with extensional well-founded relations, and whose morphisms are simulations.*

In fact, this category is a poset.

Lemma 10.1.16. *For extensional and well-founded $(A, <)$ and $(B, <)$, there is at most one simulation $f : A \to B$.*

Proof. Suppose $f, g : A \to B$ are simulations. Since being a simulation is a mere property, it suffices to show $\forall(a : A).\, (f(a) = g(a))$. By induction on $<$, we may suppose $f(a') = g(a')$ for all $a' < a$. And by extensionality of B, to have $f(a) = g(a)$ it suffices to have $\forall(b : B).\, (b < f(a)) \Leftrightarrow (b < g(a))$.

But since f is a simulation, if $b < f(a)$, then we have $a' < a$ with $f(a') = b$. By the inductive hypothesis, we have also $g(a') = b$, hence $b < g(a)$. The dual argument is symmetrical. □

Thus, if A and B are equipped with extensional and well-founded relations, we may write $A \leq B$ to mean there exists a simulation $f : A \to B$. Corollary 10.1.15 implies that if $A \leq B$ and $B \leq A$, then $A = B$.

Definition 10.1.17. An **ordinal** is a set A with an extensional well-founded relation which is *transitive*, i.e. satisfies $\forall(a, b, c : A).\, (a < b) \to (b < c) \to (a < c)$.

Example 10.1.18. Of course, the usual strict order on $\mathbb{N}$ is transitive. It is easily seen to be extensional as well; thus it is an ordinal. As usual, we denote this ordinal by ω.

Let Ord denote the type of ordinals. By the previous results, Ord is a set and has a natural partial order. We now show that Ord also admits a well-founded relation.

If A is an ordinal and $a : A$, let $A_{/a} :\equiv \{\, b : A \mid b < a \,\}$ denote the initial segment. Note that if $A_{/a} = A_{/b}$ as ordinals, then that isomorphism must respect their inclusions into A (since simulations form a poset), and hence they are equal as subsets of A. Therefore, since A is extensional, $a = b$. Thus the function $a \mapsto A_{/a}$ is an injection $A \to \mathsf{Ord}$.

Definition 10.1.19. For ordinals A and B, a simulation $f : A \to B$ is said to be **bounded** if there exists $b : B$ such that $A = B_{/b}$.

The remarks above imply that such a b is unique when it exists, so that boundedness is a mere property.

We write $A < B$ if there exists a bounded simulation from A to B. Since simulations are unique, $A < B$ is also a mere proposition.

Theorem 10.1.20. $(\mathsf{Ord}, <)$ *is an ordinal.*

More precisely, this theorem says that the type $\mathsf{Ord}_{\mathcal{U}_i}$ of ordinals in one universe is itself an ordinal in the next higher universe, i.e. $(\mathsf{Ord}_{\mathcal{U}_i}, <) : \mathsf{Ord}_{\mathcal{U}_{i+1}}$.

Proof. Let A be an ordinal; we first show that $A_{/a}$ is accessible (in Ord) for all $a : A$. By well-founded induction on A, suppose $A_{/b}$ is accessible for all $b < a$. By definition of accessibility, we must show that B is accessible in Ord for all $B < A_{/a}$. However, if $B < A_{/a}$ then there is some $b < a$ such that $B = (A_{/a})_{/b} = A_{/b}$, which is accessible by the inductive hypothesis. Thus, $A_{/a}$ is accessible for all $a : A$.

Now to show that A is accessible in Ord, by definition we must show B is accessible for all $B < A$. But as before, $B < A$ means $B = A_{/a}$ for some $a : A$, which is accessible as we just proved. Thus, Ord is well-founded.

For extensionality, suppose A and B are ordinals such that $\prod_{(C:\mathsf{Ord})}(C < A) \Leftrightarrow (C < B)$. Then for every $a : A$, since $A_{/a} < A$, we have $A_{/a} < B$, hence there is $b : B$ with $A_{/a} = B_{/b}$. Define $f : A \to B$ to take each a to the corresponding b; it is straightforward to verify that f is an isomorphism. Thus $A \cong B$, hence $A = B$ by univalence.

Finally, it is easy to see that $<$ is transitive. □

Treating Ord as an ordinal is often very convenient, but it has its pitfalls as well. For instance, consider the following lemma, where we pay attention to how universes are used.

Lemma 10.1.21. *Let $\mathcal{U}$ be a universe. For any $A : \mathsf{Ord}_{\mathcal{U}}$, there is a $B : \mathsf{Ord}_{\mathcal{U}}$ such that $A < B$.*

Proof. Let $B = A + \mathbf{1}$, with the element $\star : \mathbf{1}$ being greater than all elements of A. Then B is an ordinal and it is easy to see that $A \cong B_{/\star}$. □

The ordinal B constructed in the proof of Lemma 10.1.21 is called the **successor** of A.

This lemma illustrates a potential pitfall of the "typically ambiguous" style of using $\mathcal{U}$ to denote an arbitrary, unspecified universe. Consider the following alternative proof of it.

Another putative proof of Lemma 10.1.21. Note that $C < A$ if and only if $C = A_{/a}$ for some $a : A$. This gives an isomorphism $A \cong \mathsf{Ord}_{/A}$, so that $A < \mathsf{Ord}$. Thus we may take $B :\equiv \mathsf{Ord}$. □

The second proof would be valid if we had stated Lemma 10.1.21 in a typically ambiguous style. But the resulting lemma would be less useful, because the second proof would constrain the second "Ord" in the lemma statement to refer to a higher universe level than the first one. The first proof allows both universes to be the same.

Similar remarks apply to the next lemma, which could be proved in a less useful way by observing that $A \leq \mathsf{Ord}$ for any $A : \mathsf{Ord}$.

Lemma 10.1.22. *Let $\mathcal{U}$ be a universe. For any $X : \mathcal{U}$ and $F : X \to \mathsf{Ord}_{\mathcal{U}}$, there exists $B : \mathsf{Ord}_{\mathcal{U}}$ such that $Fx \leq B$ for all $x : X$.*

Proof. Let B be the set-quotient (see Remark 10.0.1) of the equivalence relation $\sim$ on $\sum_{(x:X)} Fx$ defined as follows:

$$(x, y) \sim (x', y') :\equiv \Big((Fx)_{/y} \cong (Fx')_{/y'} \Big).$$

Define $(x, y) < (x', y')$ if $(Fx)_{/y} < (Fx')_{/y'}$. This clearly descends to the quotient, and can be seen to make B into an ordinal. Moreover, for each $x : X$ the induced map $Fx \to B$ is a simulation. □

10.2 Classical well-orderings

We now show the equivalence of our ordinals with the more familiar classical well-orderings.

Lemma 10.2.1. *Assuming excluded middle, every ordinal is trichotomous:*

$$\forall (a, b : A).\ (a < b) \vee (a = b) \vee (b < a).$$

Proof. By induction on a, we may assume that for every $a' < a$ and every $b' : A$, we have $(a' < b') \vee (a' = b') \vee (b' < a')$. Now by induction on b, we may assume that for every $b' < b$, we have $(a < b') \vee (a = b') \vee (b' < a)$.

By excluded middle, either there merely exists a $b' < b$ such that $a < b'$, or there merely exists a $b' < b$ such that $a = b'$, or for every $b' < b$ we have $b' < a$. In the first case, merely $a < b$ by transitivity, hence $a < b$ as it is a mere proposition. Similarly, in the second case, $a < b$ by transport. Thus, suppose $\forall (b' : A).\ (b' < b) \to (b' < a)$.

Now analogously, either there merely exists $a' < a$ such that $b < a'$, or there merely exists $a' < a$ such that $a' = b$, or for every $a' < a$ we have $a' < b$. In the

first and second cases, $b < a$, so we may suppose $\forall(a' : A).\, (a' < a) \to (a' < b)$. However, by extensionality, our two suppositions now imply $a = b$. □

Lemma 10.2.2. *A well-founded relation contains no cycles, i.e.*

$$\forall(n : \mathbb{N}).\, \forall(a : \mathbb{N}_n \to A).\, \neg\Big((a_0 < a_1) \wedge \cdots \wedge (a_{n-1} < a_n) \wedge (a_n < a_0)\Big).$$

Proof. We prove by induction on $a : A$ that there is no cycle containing a. Thus, suppose by induction that for all $a' < a$, there is no cycle containing a'. But in any cycle containing a, there is some element less than a and contained in the same cycle. □

In particular, a well-founded relation must be **irreflexive**, i.e. $\neg(a < a)$ for all a.

Theorem 10.2.3. *Assuming excluded middle, $(A, <)$ is an ordinal if and only if every nonempty subset $B \subseteq A$ has a least element.*

Proof. If A is an ordinal, then by Lemma 10.1.8 every nonempty subset merely has a minimal element. But trichotomy implies that any minimal element is a least element. Moreover, least elements are unique when they exist, so merely having one is as good as having one.

Conversely, if every nonempty subset has a least element, then by Lemma 10.1.8, A is well-founded. We also have trichotomy, since for any a, b the subset $\{\, a, b \,\} :\equiv \{\, x : A \mid x = a \vee x = b \,\}$ merely has a least element, which must be either a or b. This implies transitivity, since if $a < b$ and $b < c$, then either $a = c$ or $c < a$ would produce a cycle. Similarly, it implies extensionality, for if $\forall(c : A).\, (c < a) \Leftrightarrow (c < b)$, then $a < b$ implies (letting c be a) that $a < a$, which is a cycle, and similarly if $b < a$; hence $a = b$. □

We now move on to consider consequences of the axiom of choice. For any set X, let $\mathcal{P}_+(X)$ denote the type of merely inhabited subsets of X:

$$\mathcal{P}_+(X) :\equiv \{\, Y : \mathcal{P}(X) \mid \exists(x : X).\, x \in Y \,\}.$$

Assuming excluded middle, this is equivalently the type of *nonempty* subsets of X, and we have $\mathcal{P}(X) \simeq (\mathcal{P}_+(X)) + \mathbf{1}$.

Theorem 10.2.4. *Assuming excluded middle, the following are equivalent.*

(i) For every set X, there merely exists a function $f : \mathcal{P}_+(X) \to X$ such that $f(Y) \in Y$ for all $Y : \mathcal{P}_+(X)$.
(ii) Every set merely admits the structure of an ordinal.

Proof. One direction is easy: suppose (ii). Since we aim to prove the mere proposition (i), we may assume A is an ordinal. But then we can define $f(B)$ to be the least element of B.

Now suppose (i). As before, since (ii) is a mere proposition, we may assume given such an f. We extend f to a function

$$\bar{f} : \mathcal{P}(X) \simeq (\mathcal{P}_+(X)) + \mathbf{1} \longrightarrow X + \mathbf{1}$$

in the obvious way. Now for any ordinal A, we can define $g_A : A \to X + \mathbf{1}$ by well-founded recursion:

$$g_A(a) :\equiv \bar{f}\Big(X \setminus \{\, g_A(b) \;\big|\; (b < a) \wedge (g_A(b) \in X) \,\}\Big)$$

(regarding X as a subset of $X + \mathbf{1}$ in the obvious way).

Let $A' :\equiv \{\, a : A \mid g_A(a) \in X \,\}$ be the preimage of $X \subseteq X + \mathbf{1}$; then we claim the restriction $g'_A : A' \to X$ is injective. For if $a, a' : A$ with $a \neq a'$, then by trichotomy and without loss of generality, we may assume $a' < a$. Thus $g_A(a') \in \{\, g_A(b) \mid b < a \,\}$, so since $f(Y) \in Y$ for all Y we have $g_A(a) \neq g_A(a')$.

Moreover, A' is an initial segment of A. For $g_A(a)$ lies in $\mathbf{1}$ if and only if $\{\, g_A(b) \mid b < a \,\} = X$, and if this holds then it also holds for any $a' > a$. Thus, A' is itself an ordinal.

Finally, since Ord is an ordinal, we can take $A :\equiv \mathsf{Ord}$. Let X' be the image of $g'_{\mathsf{Ord}} : \mathsf{Ord}' \to X$; then the inverse of g'_{Ord} yields an injection $H : X' \to \mathsf{Ord}$. By Lemma 10.1.22, there is an ordinal C such that $Hx \leq C$ for all $x : X'$. Then by Lemma 10.1.21, there is a further ordinal D such that $C < D$, hence $Hx < D$ for all $x : X'$. Now we have

$$\begin{aligned} g_{\mathsf{Ord}}(D) &= \bar{f}\Big(X \setminus \Big\{\, g_{\mathsf{Ord}}(B) \;\Big|\; B < D \wedge (g_{\mathsf{Ord}}(B) \in X) \,\Big\}\Big) \\ &= \bar{f}\Big(X \setminus \Big\{\, g_{\mathsf{Ord}}(B) \;\Big|\; g_{\mathsf{Ord}}(B) \in X \,\Big\}\Big) \end{aligned}$$

since if $B : \mathsf{Ord}$ and $(g_{\mathsf{Ord}}(B) \in X)$, then $B = Hx$ for some $x : X'$, hence $B < D$. Now if

$$\Big\{\, g_{\mathsf{Ord}}(B) \;\Big|\; g_{\mathsf{Ord}}(B) \in X \,\Big\}$$

is not all of X, then $g_{\mathsf{Ord}}(D)$ would lie in X but not in this subset, which would be a contradiction since D is itself a potential value for B. So this set must be all of X, and hence g'_{Ord} is surjective as well as injective. Thus, we can transport the ordinal structure on Ord' to X. □